

Table 1: Verification results for the experiments of HaliVer from Chapter 4.

Name	V	Result	Base			Qualifiers			Speedup
			#	T	σ_{mean}	#	T	σ_{mean}	
blur	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
hist	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
conv_layer	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
gemm	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	Error	10			10			
auto_viz	0	✓	10	1	1	10	1	1	1.0
	1	✓	9	1	1	10	1	1	1.0
		✗	1	1	1	0			
	2	✓	10	1	1	10	1	1	1.0
bilateral_grid	3	✓	10	1	1	10	1	1	1.0
		✓	10	1	1	10	1	1	1.0
		✓	10	1	1	10	1	1	1.0
		✓	10	1	1	10	1	1	1.0
Total				22	5		22	5	1.0

Table 2: Verification results for the experiments of HaliVer from Chapter 4.

Name	V	Result	Base			Qualifiers			Speedup
			#	T	σ_{mean}	#	T	σ_{mean}	
blur	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
hist	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
conv_layer	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
gemm	0	✓	10	1	1	10	1	1	1.0
	1	✓	10	1	1	10	1	1	1.0
	2	✓	10	1	1	10	1	1	1.0
	3	Error	10			10			
auto_viz	0	✓	10	1	1	9	1	1	1.0
		✗	0			1	1	1	
	1	✓	8	1	1	10	1	1	1.0
		✗	2	1	1	0			
	2	✓	10	1	1	10	1	1	1.0
	3	✓	10	1	1	10	1	1	1.0
Total			19			19			1.0

Table 3: Verification results for `step`, `sub_direction`, `solve_direction`, and `perform_iteration` produced by HaliVer. We use abbreviations for versions with concrete bounds (**CB**), nonconcrete bounds (**NCB**), **unique** and **const** type qualifiers, and no type qualifiers (**Normal**).

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	10	1	1	10	1	1	1.0
NCB	✓	10	1	1	10	1	1	1.0

(a) `step`

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	10	1	1	10	1	1	1.0
NCB	✓	0			1	1	1	1.0
	✗	10	1	1	9	1	1	

(b) `sub_direction`

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	0			10	1	1	1.0
	✗	2	1	1	0			
	T.O.	8	-	-	0			
NCB	✗	9	1	1	6	1	1	
	T.O.	1	-	-	4	-	-	

(c) `solve_direction`

Version	Result	Base			Qualifiers			Speedup
		#	T	σ_{mean}	#	T	σ_{mean}	
CB	✓	0			10	1	1	1.0
	✗	6	1	1	0			
	T.O.	4	-	-	0			
NCB	✗	1	1	1	0			
	T.O.	9	-	-	10	-	-	

(d) `perform_iteration`